

Markov Chain

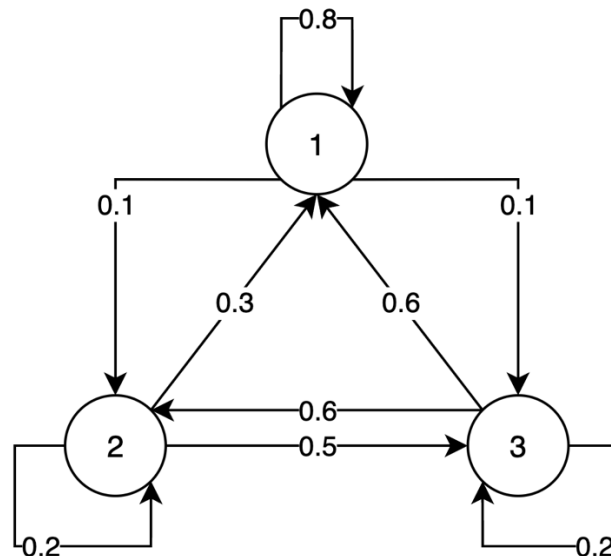
假设我们要做一个谷歌地图，我们只有三条高速公路，我们要预测的是现在各个高速上行驶的车辆，最终会去往哪个高速，这样我们可以预测出来，哪个高速将要发生堵车，可以保证用户有更好的使用体验。

从第一条高速公路出发，依然维持在第一条的概率是 0.8

从第一条高速公路出发，转向第二条高速公路的概率是 0.1

从第一条高速公路出发，转向第三条高速公路的概率是 0.1

...



那么我们可以将上述改变成一个矩阵：

$$\begin{pmatrix} 0.8 & 0.3 & 0.2 \\ 0.1 & 0.2 & 0.6 \\ 0.1 & 0.5 & 0.2 \end{pmatrix}$$

在这个矩阵中 a_{ij} 的意思为状态 i 转移到状态 j 的概率。

Example:

By reviewing its donation records, the alumni office of a college finds that 80% of its alumni who contribute to the annual fund one year will also contribute the next year, and 30% of those who do not contribute one year will contribute the next.

某大学校友办公室通过查阅捐赠记录发现，该校 80%的校友在某一年捐赠了年度基金，第二年也会捐赠；30%的校友在某一年没有捐赠，第二年也会捐赠。

This can be viewed as a Markov chain with two states:

1. state 1 corresponds to an alumnus giving a donation in any one year,
2. and state 2 corresponds to the alumnus not giving a donation in that year.

这可以看作是一个具有两个状态的马尔可夫链：

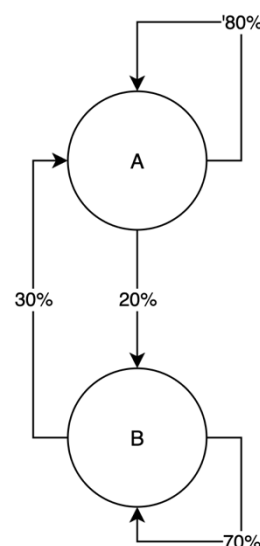
1. 状态 1 对应于校友在某一年捐赠；
2. 状态 2 对应于校友在当年没有捐赠。

The transition matrix is?

转移矩阵是什么？

$$P = \begin{pmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{pmatrix}$$

- 如果某一年捐赠，则下一年有 80%的概率继续捐赠；
- 从**捐赠**到**不捐赠**的转移概率为 0.2；
- 从**不捐赠**到**捐赠**的转移概率为 0.3；
- 表示如果某一年未捐赠，则下一年有 70%的概率继续不捐赠。



Note:

在这种矩阵中，每一列的总和一定为 1，这是因为如果这个系统只有 k 个 state，那么任意一个 state 最终一定会转移到这 k 个 state 中

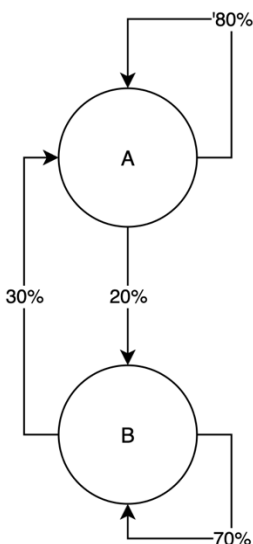
这样的矩阵我们称之为 stochastic matrix, a probability matrix, or a Markov matrix.

Markov matrix

If a Markov chain has k possible states, which we label as $1, 2, \dots, k$, then the probability that the system is in state i at any observation after it was in state j at the preceding observation is denoted by p_{ij} and is called the transition probability from state j to state i .

The matrix $P = [p_{ij}]$ is called the transition matrix of the Markov chain.

在状态转移矩阵中，第 j 列，第 i 行的数字，代表着他从状态 j 变化到状态 i 的概率



$$P = \begin{pmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{pmatrix}$$

比如 P_{11} 表示从状态一变化到状态一的概率

比如 P_{21} 表示从状态一变化到状态二的概率

State Vector

State Vector

The state vector for an observation of a Markov chain with k states is a column vector x whose i th component x_i is the probability that the system is in the i th state at that time.

状态向量 (State Vector) 是用于描述系统在某一时刻处于各个状态的**概率分布**的列向量。

Example:

假设有一个交通信号灯的模型，它有三个状态

- S_1 : 绿灯
- S_2 : 黄灯
- S_3 : 红灯

如果系统在某一时刻的状态向量为

$$x(t) = \begin{pmatrix} 0.6 \\ 0.1 \\ 0.3 \end{pmatrix}$$

这就意味着在该时刻，交通信号灯显示绿灯的概率为 60%，黄灯的概率为 10%，红灯的概率为 30%。

Example:

如前面提到的校友捐赠模型，有两个状态：

- S_1 : 捐赠
- S_2 : 不捐赠

如果某年的状态向量为

$$x(t) = \begin{pmatrix} 0.5 \\ 0.5 \end{pmatrix}$$

则表示那一年捐赠与不捐赠的概率各占 50%。

假设一个 Markov 链有 **k 个状态**，分别记作：

$$S_1, S_2, \dots, S_k$$

某一时刻，系统在每个状态的**概率**可以用一个列向量表示，这个向量被称为**状态向量 (State Vector)**。

状态向量通常记作：

$$x(t) = \begin{pmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_k(t) \end{pmatrix}$$

$x_i(t)$ 表示系统在时刻 t 处于状态 S_i 的概率

Example

For the Transition Matrix

$$P = \begin{bmatrix} 0.8 & 0.3 & 0.2 \\ 0.1 & 0.2 & 0.6 \\ 0.1 & 0.5 & 0.2 \end{bmatrix}$$

假设一辆车从第二条高速进入，那么它现在的 State vector 是什么？

$$x^{(0)} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

因为他现在已经确定了自己是在第二条高速上

那么它有可能去到的下一条高速公路的概率是？

$$x^{(1)} = \begin{bmatrix} 0.3 \\ 0.2 \\ 0.5 \end{bmatrix} = \begin{bmatrix} 0.8 & 0.3 & 0.2 \\ 0.1 & 0.2 & 0.6 \\ 0.1 & 0.5 & 0.2 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

解释

$$0.3 = 0.8 * 0 + 0.3 * 1 + 0.2 * 0$$

[0.8 0.3 0.2]中的 0.3 意味着：从第二个状态到第一个个状态的概率（从第二条高速到第一条高速的概率）

这里的 0.3, 0.2, 0.5 意味着什么？

0.3 意味着：

0.2 意味着：

0.5 意味着：

那么它可能去到的再下一条高速公路的概率是？

它去一号高速公路的概率是：

$$0.8 * 0.3 + 0.3 * 0.2 + 0.2 * 0.5 = 0.4$$

它去二号高速公路的概率是

$$0.3 * 0.3 + 0.2 * 0.2 + 0.5 * 0.5 = 0.37$$

它去三号高速公路的概率是

$$0.2 * 0.3 + 0.6 * 0.2 + 0.2 * 0.5 = 0.23$$

因此，我们最终可以写成这样的式子

$$x^{(2)} = \begin{bmatrix} 0.4 \\ 0.37 \\ 0.23 \end{bmatrix} = \begin{bmatrix} 0.8 & 0.3 & 0.2 \\ 0.1 & 0.2 & 0.6 \\ 0.1 & 0.5 & 0.2 \end{bmatrix} \begin{bmatrix} 0.3 \\ 0.2 \\ 0.5 \end{bmatrix}$$

因此我们可以得出一个结论：

$$x^{(n+1)} = Px^{(n)}$$

这里的 P 是状态转移矩阵

Theorem

If P is the transition matrix of a Markov chain and $x(n)$ is the state vector at the nth observation, then $x^{(n+1)} = Px^{(n)}$.

Exercise:

我们的状态转移矩阵是：

$$P = \begin{bmatrix} 0.8 & 0.3 & 0.2 \\ 0.1 & 0.2 & 0.6 \\ 0.1 & 0.5 & 0.2 \end{bmatrix}$$

TABLE 1												
$x^{(n)} \backslash n$	0	1	2	3	4	5	6	7	8	9	10	11
$x_1^{(n)}$	0	.300	.400		.511	.533	.544		.553	.555		.557
$x_2^{(n)}$	1	.200	.370		.261	.240	.238		.232	.231		.230
$x_3^{(n)}$	0	.500	.230		.228	.227	.219		.215	.214		.213

- 1. 计算 $x^{(3)}$
- 2. $x^{(6)}$ 的意义是什么
- 3. 计算 $x^{(7)}$
- 4. 计算 $x^{(10)}$

Answer:

TABLE 1												
$x^{(n)} \backslash n$	0	1	2	3	4	5	6	7	8	9	10	11
$x_1^{(n)}$	0	.300	.400	.477	.511	.533	.544	.550	.553	.555	.556	.557
$x_2^{(n)}$	1	.200	.370	.252	.261	.240	.238	.233	.232	.231	.230	.230
$x_3^{(n)}$	0	.500	.230	.271	.228	.227	.219	.217	.215	.214	.214	.213

Exercise:

A traffic officer is assigned to control the traffic at the eight intersections indicated in Figure below. She is instructed to remain at each intersection for an hour and then to either remain at the same intersection or move to a neighboring intersection. To avoid establishing a pattern, she is told to choose her new intersection on a random basis, with each possible choice equally likely.

一名交通警察被指派控制下图所示八个路口的交通。她被指示在每个路口停留一小时，然后要么留在同一个路口，要么移至相邻的路口。为了避免形成模式，她被要求随机选择新的路口，每个可能的选择都有同等的可能性。

For example, if she is at intersection 5, her next intersection can be 2, 4, 5, or 8, each with probability .

例如，如果她在路口 5，她的下一个路口可能是 2、4、5 或 8，每个路口的概率为 $\frac{1}{4}$ 。因此，在状态转移矩阵里

$$P_{52} = \frac{1}{4}, P_{58} = \frac{1}{4}, P_{55} = \frac{1}{4}, P_{54} = \frac{1}{4}$$

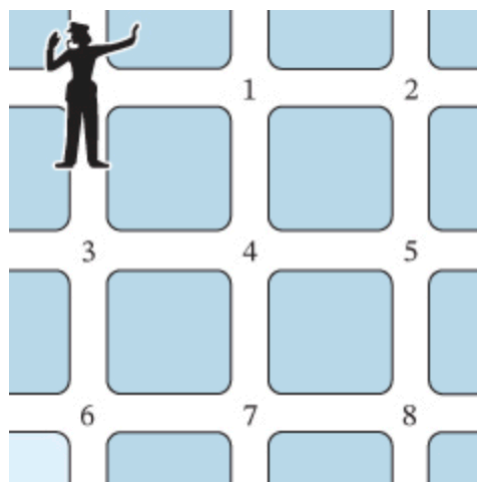
因此，在状态转移矩阵里

$$P_{52} = \frac{1}{4}, P_{58} = \frac{1}{4}, P_{55} = \frac{1}{4}, P_{54} = \frac{1}{4}$$

例如，如果她在路口 4，她的下一个路口可能是 1、3、4、5 或 7，每个路口的概率为 $\frac{1}{5}$ 。

这个马尔可夫链的转移矩阵是

Old Intersection								
1	2	3	4	5	6	7	8	
$\frac{1}{3}$	$\frac{1}{3}$	0	$\frac{1}{5}$	0	0	0	0	1
$\frac{1}{3}$	$\frac{1}{3}$	0	0	$\frac{1}{4}$	0	0	0	2
0	0	$\frac{1}{3}$	$\frac{1}{5}$	0	$\frac{1}{3}$	0	0	3
$\frac{1}{3}$	0	$\frac{1}{3}$	$\frac{1}{5}$	$\frac{1}{4}$	0	$\frac{1}{4}$	0	4
0	$\frac{1}{3}$	0	$\frac{1}{5}$	$\frac{1}{4}$	0	0	$\frac{1}{3}$	5
0	0	$\frac{1}{3}$	0	0	$\frac{1}{3}$	$\frac{1}{4}$	0	6
0	0	0	$\frac{1}{5}$	0	$\frac{1}{3}$	$\frac{1}{4}$	$\frac{1}{3}$	7
0	0	0	0	$\frac{1}{4}$	0	$\frac{1}{4}$	$\frac{1}{3}$	8



如果交通警察从 5 号交叉路口出发，那么她每小时的位置的概率是什么？填写下表：

$x^{(n)}$ n	0	1	2	3	4
$x_1^{(n)}$					
$x_2^{(n)}$					
$x_3^{(n)}$					
$x_4^{(n)}$					
$x_5^{(n)}$					
$x_6^{(n)}$					
$x_7^{(n)}$					
$x_8^{(n)}$					

Answer:

$x^{(n)}$ \ n	0	1	2	3	4
$x_1^{(n)}$	0	.000	.133	.116	.130
$x_2^{(n)}$	0	.250	.146	.163	.140
$x_3^{(n)}$	0	.000	.050	.039	.067
$x_4^{(n)}$	0	.250	.113	.187	.162
$x_5^{(n)}$	1	.250	.279	.190	.190
$x_6^{(n)}$	0	.000	.000	.050	.056
$x_7^{(n)}$	0	.000	.133	.104	.131
$x_8^{(n)}$	0	.250	.146	.152	.124

要点:

第一次状态转移:

$$x^{(1)} = Px^{(0)}$$

第二次状态转移:

$$x^{(2)} = Px^1 = P(Px^0) = P^2x^{(0)}$$

第三次状态转移:

$$x^{(3)} = Px^2 = P(P^2x^0) = P^3x^{(0)}$$

...

第 n 次状态转移:

$$x^{(n)} = P^n x^{(0)}$$

True or false:

是否对于一个 state $x^{(0)}$, 经过了很多转移之后, 它一定会趋近于一个固定的 vector 呢?

这里举一个这样的例子

$$\begin{pmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{pmatrix}$$

Answer:

不一定, 考虑

$$P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, x^{(0)} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$x^{(1)} = Px^{(0)} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$x^{(2)} = Px^{(1)} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$x^{(3)} = Px^{(2)} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

...

Regular Matrix

Regular

A transition matrix is regular if some integer power of it has all positive entries.

存在一个 m ，可以使 P^m 中的每一个数都为正数，那么 matrix P 是 regular 的

假设有以下转移矩阵 P ：

$$P = \begin{bmatrix} 0.2 & 0.5 \\ 0.8 & 0.5 \end{bmatrix}$$

那么 P^2 为：

$$P^2 = \begin{bmatrix} 0.44 & 0.35 \\ 0.56 & 0.65 \end{bmatrix}$$

但是如果有以下转移矩阵 P ：

$$P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

那么 P^2 为：

$$P^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

那么 P^3 为：

$$P^3 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

那么不管怎么样，一定不存在一个 m ，使得 P^m 中的每一项都为正数。因此， P 不是 regular 的

Example

展示 P is not regular

$$P^n = \begin{bmatrix} \frac{1}{2} & 0 \\ \frac{1}{2} & 1 \end{bmatrix}$$

$$P^n = \begin{bmatrix} \left(\frac{1}{2}\right)^n & 0 \\ 1 - \left(\frac{1}{2}\right)^n & 1 \end{bmatrix}, n = 1, 2, \dots \text{ Thus, no integer power of } P \text{ has all positive entries.}$$

Show that the transition matrix is regular

$$P = \begin{bmatrix} 0 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} \end{bmatrix}$$

$$P^2 = \begin{bmatrix} \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{2} \end{bmatrix}$$

Regular Markov Chain

A Markov chain that is governed by a regular transition matrix is called a regular Markov chain. We will see that every regular Markov chain has a fixed state vector q such that $P^n x^{(0)}$ approaches q as n increases for any choice of $x^{(0)}$

无论你最开始处于哪种状态，经过足够多次的随机“转移”，只要你的转移矩阵是 regular 的，最终都会达到一个固定的状态分布 q

这个东西的证明十分阴间，涉及到很多我们没学过的知识，因此我们不会在这里证明这个玩意，但是我们需要知道的是下面这个定理

Theorem

If P is a regular transition matrix, then as $n \rightarrow \infty$,

$$P^n \rightarrow \begin{bmatrix} q_1 & q_1 & \cdots & q_1 \\ q_2 & q_2 & \cdots & q_2 \\ \vdots & \vdots & \ddots & \vdots \\ q_k & q_k & \cdots & q_k \end{bmatrix}$$

Where $q_1 + q_2 + \cdots + q_k = 1$

那么当他收敛到 q ，意味着 $x^n = q$ ， $x^{n+1} = q$ ，一定不会变，为什么

$$\begin{aligned} Q &= \begin{bmatrix} q_1 & q_1 & \cdots & q_1 \\ q_2 & q_2 & \cdots & q_2 \\ \vdots & \vdots & \ddots & \vdots \\ q_k & q_k & \cdots & q_k \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_k \end{bmatrix} \\ &= \begin{bmatrix} q_1 x_1 + q_1 x_2 + \cdots + q_1 x_k \\ q_2 x_1 + q_2 x_2 + \cdots + q_2 x_k \\ \vdots \\ q_k x_1 + q_k x_2 + \cdots + q_k x_k \end{bmatrix} \\ &= (x_1 + x_2 + \cdots + x_k) \begin{bmatrix} q_1 \\ q_2 \\ \vdots \\ q_k \end{bmatrix} \\ &= 1 \begin{bmatrix} q_1 \\ q_2 \\ \vdots \\ q_k \end{bmatrix} \\ &= q \end{aligned}$$

确实，再经过状态转移矩阵之后，我们的 state vector 没有任何改变，与定理吻合

Theorem

If P is a regular transition matrix and x is any probability vector, then as $n \rightarrow \infty$,

$$P^n x \rightarrow \begin{bmatrix} q_1 \\ q_2 \\ \vdots \\ q_k \end{bmatrix} = q$$

Where q is a fixed probability vector, independent of n , all of whose entries are positive

Steady-State Vector

The steady-state vector q of a regular transition matrix P is the unique probability vector that satisfies the equation $Pq = q$.

当然，大多数时候，我们会这么写这个式子：

$$(I - P)q = 0$$

Example

找到下述转移矩阵的 Steady State Vector

$$\begin{pmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{pmatrix}$$

The linear system $(I - P)q = 0$ 就可以写为：

$$\left(\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - \begin{pmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{pmatrix} \right) \begin{pmatrix} q_1 \\ q_2 \end{pmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{pmatrix} 0.2 & -0.3 \\ -0.2 & 0.3 \end{pmatrix} \begin{pmatrix} q_1 \\ q_2 \end{pmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

解上式可得：

$$\begin{aligned} 0.2q_1 - 0.3q_2 &= 0 \\ -0.2q_1 + 0.3q_2 &= 0 \end{aligned}$$

因此：

$$q_1 = 1.5q_2$$

又因为 $q_1 + q_2 = 1$, 因此：

$$\begin{cases} q_1 = 0.6 \\ q_2 = 0.4 \end{cases}$$

Exercise:

找到下述转移矩阵的 Steady State Vector

$$P = \begin{bmatrix} 0.8 & 0.3 & 0.2 \\ 0.1 & 0.2 & 0.6 \\ 0.1 & 0.5 & 0.2 \end{bmatrix}$$

Example

John is either happy or sad. If he is happy one day, then he is happy the next day four times out of five. If he is sad one day, then he is sad the next day one time out of three. Over the long term, what are the chances that John is happy on any given day?

Let $\mathbf{x} = [x_1 \ x_2]^T$ be the state vector, with x_1 = probability that John is happy and x_2 = probability that John is sad. The transition matrix P will be $P = \begin{bmatrix} \frac{4}{5} & \frac{2}{3} \\ \frac{1}{5} & \frac{1}{3} \end{bmatrix}$ since the columns must sum to one. We find the steady state vector for P by solving $\begin{bmatrix} \frac{1}{5} & -\frac{2}{3} \\ -\frac{1}{5} & \frac{2}{3} \end{bmatrix} \begin{bmatrix} q_1 \\ q_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, i.e., $\frac{1}{5}q_1 = \frac{2}{3}q_2$, so $\mathbf{q} = \begin{bmatrix} \frac{10}{3} \\ 1 \end{bmatrix} s$. Let $s = \frac{3}{13}$ and get $\mathbf{q} = \begin{bmatrix} \frac{10}{13} \\ \frac{3}{13} \end{bmatrix}$, so $\frac{10}{13}$ is the probability that John will be happy on a given day.

Example

Rasputin and Charlemagne are playing a zero sum game. The payoff matrix for Rasputin (the row player) is below, where each entry is in units of dollars (\$).

普京和查理曼大帝正在玩一场零和游戏。普京（行玩家）的收益矩阵如下，其中每个条目都以美元 (\$) 为单位。

$$\begin{pmatrix} 1 & -3 & 5 \\ 2 & -1 & 2 \\ -4 & 0 & 6 \\ -2 & 6 & -1 \end{pmatrix}$$

After playing the game for a while, Rasputin has observed that Charlemagne plays move 1 half the time, move 2 half the time, and he never plays move 3. Based on this, what strategy would you recommend to Rasputin, to maximize his expected payoff? If he follows your advice, what will Rasputin's expected payoff be? Show your reasoning and then fill in the blanks below.

玩了一段时间后，普京发现查理曼大帝

- 一半时间会移动到 1 ，
- 一半时间会移动到 2 ，
- 而他从不动移到 3 。

基于此，你会向拉斯普京推荐什么策略来最大化他的预期收益？如果他听从你的建议，拉斯普京的预期收益是多少？请说明你的理由，然后填写下面的空白。

Charlemagne's strategy is $\mathbf{q} = \begin{pmatrix} 1/2 \\ 1/2 \\ 0 \end{pmatrix}$.

Let's represent Rasputin's strategy by the vector $\mathbf{p} = (p_1 \ p_2 \ p_3 \ p_4)$, where each p_i represents a chance between 0 and 1 (inclusive) and $p_1 + p_2 + p_3 + p_4 = 1$. We want to maximize his expected payoff,

$$\mathbf{p}A\mathbf{q} = (p_1 \ p_2 \ p_3 \ p_4) \begin{pmatrix} 1 & -3 & 5 \\ 2 & -1 & 2 \\ -4 & 0 & 6 \\ -2 & 6 & -1 \end{pmatrix} \begin{pmatrix} 1/2 \\ 1/2 \\ 0 \end{pmatrix} = (p_1 \ p_2 \ p_3 \ p_4) \begin{pmatrix} -1 \\ 1/2 \\ -2 \\ 2 \end{pmatrix}.$$

From here, we can see that Rasputin's expected payoff will be largest if $p_1 = p_2 = p_3 = 0$ and $p_4 = 1$. In other words, Rasputin should always play **move 4**. Thus $(0 \ 0 \ 0 \ 1)$ is the strategy we advise him to use. If he follows our advice, his expected payoff per game is $\mathbf{p}A\mathbf{q} = 2$.

I would advise Rasputin to use the strategy $\mathbf{p} = \left(\boxed{0} \ \boxed{0} \ \boxed{0} \ \boxed{1} \right)$.

With this strategy, Rasputin's expected payoff per game is 2.

Exercise: 回家练习

找到下述转移矩阵的 Steady State Vector

$$P = \begin{bmatrix} 0 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} \end{bmatrix}$$

A country is divided into three demographic regions. It is found that:

1. each year 5% of the residents of region 1 move to region 2, and 5% move to region 3.
2. Of the residents of region 2, 15% move to region 1 and 10% move to region 3.
3. And of the residents of region 3, 10% move to region 1 and 5% move to region

What percentage of the population resides in each of the three regions after a long period of time?

Answer:

$54\frac{1}{6}$ in region 1, $16\frac{2}{3}$ in region 2, $29\frac{1}{6}$ in region 3